Dear interviewers:

Thank you for taking the time to review my report on question 1, "**Application for Lending Department of a Bank Part 2**". You can refer to more details in my github repo.

Yours sincerely,
Lixin Tu

Application for Lending Department of a Bank

1 Part 1. Modeling of a balance sheet

Please find the answer for part 1 in our previous submission.

2 Part 2: Consider the problem of applying large language model to financial statement analysis, such as what has been done in Alonso(24), Farr(25), and Zhang(25).

Comment 2.1

a) Choose your favourite LLM to apply the problem of financial statement analysis.

Response:

I selected DeepSeek R1 (deepseek-r1-250528) because of its strong reasoning capabilities (Chain of Thought). Financial statement analysis requires complex logical deduction and numerical understanding, and R1's architecture is optimized to handle such multi-step reasoning tasks effectively.

Comment 2.2

b) Let's try the task of balance sheet forecast using the same set of data as collected in part 1, does the LLM you picked perform better or worse than your model?

Response:

The Part 1 model (MLP + Constraint) performs significantly better than the LLM-picked linear regression model, achieving a much lower Mean Squared Error (MSE) of 312.20 compared to 920.12. This performance advantage is likely due to the use of a neural network in the Part 1 model, which enables it to capture non-linear relationships in the data, as well as the explicit enforcement of the accounting identity Assets $=$ Liabilities $+$ Equity through a custom constraint layer. In contrast, the linear regression model selected by the LLM lacks both the representational capacity to model complex structural dependencies and the ability to incorporate domain-specific constraints, resulting in inferior predictive performance.

Comment 2.3

c) Is it possible to combine your model in part 1 and LLM to create an ensemble model that performs better than the individual model in balance sheet forecast?

Response:

We introduced a new analysis script called `run_improved_agent.py`, which directed the agent to adopt non-linear models (Random Forest and Gradient Boosting) instead of simple Linear Regression. Key improvements included handling the multi-output nature of the problem and implementing post-processing to enforce the fundamental accounting identity:

$$\text{Assets} = \text{Liabilities} + \text{Equity} \tag{1}$$

The performance comparison regarding Mean Squared Error (MSE) is as follows:

- Previous LLM Model (Linear Regression): MSE ≈ 920

- Baseline Part 1 Model (Neural Network): MSE ≈ 312
- **New Agent Model (Random Forest): MSE** ≈ 203.44

By properly utilizing the agent with optimizing its environment and providing expert instructions, we achieved a model that outperforms both the original Part 1 model and the naive LLM linear regression. The agent-trained Random Forest regressor reduced the error by approximately 35% compared to the baseline.

Comment 2.4

d) Given the results of your analysis, pick a company you have analysed, what would you recommend to the CFO or CEO of this particular company given your results?

Response:

Based on our predictive modeling using the Random Forest algorithm, which demonstrated superior accuracy with a Mean Squared Error of 203.44 (significantly outperforming the gradient boosting baseline), I have selected JPMorgan Chase for this strategic recommendation. The model forecasts JPMorgan's retained earnings for the fifth year to reach 36.74, the highest among the seven analyzed entities and distinctly surpassing the technology sector average of approximately 30.6 (observed in Microsoft and Tencent). Furthermore, our descriptive statistics revealed a significant standard deviation of 15.96 for cash equivalents across the dataset, indicating high variability in corporate liquidity strategies. JPMorgan's position at the extreme upper end of this spectrum suggests an ultra-conservative approach to capital preservation that may be exceeding necessary risk buffers.

Therefore, I recommend that the CFO evaluate this projected surplus of 36.74 against the firm's cost of capital. To prevent the dilution of Return on Equity (ROE) associated with such high retention, the leadership should consider deploying the capital exceeding the 35.0 threshold into share buybacks or increased dividends, thereby optimizing capital efficiency while maintaining the financial stability required by the banking sector.

Comment 2.5

e) Consider the annual report of General Motors here. https://investor.gm.com/static-files/1fff6f59-551f-4fe0-bca9-74bfc9a56aeb on page 56 you can the income statement, and on page 57, the balance sheet.

Response:

We extracted the Consolidated Income Statements and Consolidated Balance Sheets from the General Motors 2023 Annual Report (PDF) and attached them as Table 1 and Table 2.

Table 1: General Motors Consolidated Income Statements (in millions, except per share amounts)

	Years Ended December 31,		
	2023	**2022**	**2021**
Net sales and revenue			
Automotive	$157,658	$143,975	$113,590
GM Financial	14,184	12,760	13,414
Total net sales and revenue	171,842	156,735	127,004
Costs and expenses			
Automotive and other cost of sales	141,330	126,892	100,544
GM Financial interest, operating and other expenses	11,374	8,862	8,582
Automotive and other selling, general and admin expense	9,840	10,667	8,554
Total costs and expenses	162,544	146,421	117,680
Operating income (loss)	9,298	10,315	9,324
Automotive interest expense	911	987	950
Interest income and other non-operating income, net	1,537	1,432	3,041
Equity income (loss)	480	837	1,301
Income (loss) before income taxes	10,403	11,597	12,716

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Table 1: General Motors Consolidated Income Statements (Continued)

	Years Ended December 31,		
	2023	**2022**	**2021**
Income tax expense (benefit)	563	1,888	2,771
Net income (loss)	9,840	9,708	9,945
Net loss (income) attributable to noncontrolling interests	287	226	74
Net income (loss) attributable to stockholders	$10,127	$9,934	$10,019
Net income (loss) attributable to common stockholders	$10,022	$8,915	$9,837
Earnings per share			
Basic earnings per common share	$7.35	$6.17	$6.78
Diluted earnings per common share	$7.32	$6.13	$6.70
Weighted-average common shares outstanding			
Basic	1,364	1,445	1,451
Diluted	1,369	1,454	1,468

Table 2: General Motors Consolidated Balance Sheet (in millions)

	Dec 31, 2023	**Dec 31, 2022**
ASSETS		
Current Assets		
Cash and cash equivalents	$18,853	$19,153
Marketable debt securities	7,613	12,150
Accounts and notes receivable, net	12,378	13,333
GM Financial receivables, net	39,076	33,623
Inventories	16,461	15,366
Other current assets	7,238	6,825
Total current assets	101,618	100,451
Non-current Assets		
GM Financial receivables, net	45,043	40,591
Equity in net assets of nonconsolidated affiliates	10,613	10,176
Property, net	50,321	45,248
Goodwill and intangible assets, net	4,862	4,945
Equipment on operating leases, net	30,582	32,701
Deferred income taxes	22,339	20,539
Other assets	7,686	9,386
Total non-current assets	171,446	163,586
Total Assets	$273,064	$264,037
LIABILITIES AND EQUITY		
Current Liabilities		
Accounts payable	$28,114	$27,486
Short-term debt and current portion of long-term debt:		
Automotive	428	1,959
GM Financial	38,540	36,819
Accrued liabilities	27,364	24,910
Total current liabilities	94,445	91,173
Non-current Liabilities		
Long-term debt:		
Automotive	15,985	15,885
GM Financial	66,788	60,036
Postretirement benefits other than pensions	4,345	4,193
Pensions	6,680	5,698

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Table 2: General Motors Consolidated Balance Sheet (Continued)

	Dec 31, 2023	Dec 31, 2022
Other liabilities	16,515	14,767
Total non-current liabilities	110,312	100,579
Total Liabilities	204,757	191,752
Equity		
Noncontrolling interest - Cruise stock incentive	118	357
Common stock	12	14
Additional paid-in capital	19,130	26,428
Retained earnings	55,391	49,251
Accumulated other comprehensive loss	(10,247)	(7,901)
Total stockholders' equity	64,286	67,792
Noncontrolling interests	3,903	4,135
Total Equity	68,189	71,927
Total Liabilities and Equity	$273,064	$264,037

Comment 2.6

f) Given this PDF file, use any tool you can think of, including any LLM, or any other tools such as chatPDF, pypdf, IronPDF, build an automatic tool to extract the income statement and balance sheet to answer the following questions: i. What's the net income of the current year? ii. What's the cost-to-income ratio? iii. What are the quick ratio, debt-to-equity ratio, debt to assets ratio, debt-to-capital ratio, debt-to-EBITDA ratio? iv. What's the interest coverage ratio?

Response:

Table 3 demonstrated the extracted results with the LLM tool deepseek-r1-250528.

Table 3: Summary of Key Financial Figures (2023)

Metric	**2023 Value**
Profitability Metrics	
Net Income	$10,127 million
Cost-to-Income Ratio	94.6%
Liquidity Metrics	
Quick Ratio	0.90
Solvency & Leverage Metrics	
Debt-to-Equity Ratio	1.79
Debt-to-Assets Ratio	44.6%
Debt-to-Capital Ratio	64.1%
Debt-to-EBITDA Ratio (proxy)	13.10
Interest Coverage Ratio	10.21

Comment 2.7

g) if you are using LLM, is the model output robust to different runs? And please provide the model API version you use, such as ’gpt-4o-2024-08-06’ for our testing. If you use other tools, please state clearly what you do and the version of the tools.

Response:

deepseek-r1-250528